CAPSTONE PROJECT · AI / ML · 2026

AI-Based Fake Identity & Document Screening System

AI-Powered Document Intelligence,
Forgery Detection & Identity Consistency Screening

AUTHOR	Sumit Kumar Barnwal
TYPE	6-Month Major Capstone Project
STATUS	In Development — Planning Phase
BRANCH	main
COMMIT	a98c21d
REPO	github.com/sumitkumarbarnwal/...

Python 3.10+ | FastAPI | OpenCV | Tesseract OCR | Machine Learning | PyTorch | PostgreSQL | Docker

A 6-month Major Capstone Project combining OCR, computer vision, image forensics, machine learning, anomaly detection, and explainable risk assessment to support preliminary screening of identity documents.

> **Repository Status:** This repository was initialized on 17 September 2026 and is currently in the early development and planning phase. All features described in this document are **planned** unless explicitly marked otherwise. This README accurately documents the proposed architecture, verified research foundation, and development roadmap.

Table of Contents

- Project Overview
- Problem Statement
- Project Objectives
- Key Features
- System Workflow
- Important Fraud Scenario
- Types of Anomalies
- AI / ML Architecture
- Research and Literature Foundation
- Research Gap
- Dataset Strategy
- System Architecture
- Technology Stack
- Project Structure
- Installation
- Environment Variables
- API Documentation
- Database Design
- Screenshots
- Testing Strategy
- Model Evaluation
- Security and Privacy
- Limitations
- Roadmap
- Responsible AI Disclaimer
- Contributing
- License
- Author

1. Project Overview

The **AI-Based Fake Identity and Document Screening System** is a 6-month capstone project designed to explore and implement an AI-driven pipeline for the preliminary screening of identity documents. The proposed

system aims to detect potential manipulation, forgery, or identity inconsistencies in submitted documents by combining multiple complementary techniques:

Technique	Role
OCR (Optical Character Recognition)	Extract structured text fields from document images
Document Classification	Identify the type of identity document
Image Preprocessing	Normalize, denoise, and prepare images for downstream analysis
Computer Vision	Analyse visual structure, layout, and appearance of documents
Image Forensics	Detect digital manipulation artefacts such as copy-paste regions, splicing, or re-compression
Machine Learning	Classify documents and detect anomalous patterns
Field Validation	Cross-validate extracted text fields against expected formats and rules
Duplicate Detection	Identify whether the same document or document number has been seen before
Identity Consistency Checking	Flag cases where a known document number appears with different identity fields
Anomaly Detection	Generate structured anomaly signals for suspicious indicators
Risk Assessment	Compute a composite risk score based on detected anomalies
Explainable AI	Communicate the evidence and reasoning behind every risk signal

> **Important:** This system is designed as a **screening and decision-support tool**, not as an authoritative mechanism for declaring a document or person fraudulent. All high-risk signals are intended to guide human review, not to replace it.

2. Problem Statement

Identity documents are critical instruments of trust used for onboarding into financial services, government processes, healthcare, education, and employment. As a consequence, they are also targets for manipulation and forgery.

Categories of Potential Manipulation

Category	Examples
Field-level text editing	Name modification, date-of-birth change, document number alteration
Photograph manipulation	Photograph replacement, face splicing

Category	Examples
Copy-paste and splicing	Inserting content from another document
Font and layout modification	Replacing text with a visually similar but different font
Text removal	Erasing original text and re-entering different values
Recapture / reprint	Printing a digitally modified document and re-photographing it
Screenshot substitution	Submitting a screenshot of a document rather than the original
Synthetic / AI-generated documents	Fully fabricated documents generated using image synthesis tools
Duplicate documents	Submitting the same document number for multiple identities
Identity-field inconsistency	Same document number associated with different names or dates of birth

Why OCR Alone Is Insufficient

A naive OCR-based approach can extract text fields from a document image:

```
Name : Rahul Kumar
Document No. : ABC1234567
Date of Birth : 12-05-2001
```

However, OCR alone **cannot** determine:

- Whether the extracted text has been digitally modified after the document was issued
- Whether the photograph on the document belongs to the person presenting it
- Whether the document's visual layout is consistent with a genuine specimen
- Whether the document number has previously been associated with a different identity
- Whether the image contains forensic artefacts indicating digital manipulation

Addressing these gaps requires a multi-modal pipeline combining text extraction, visual analysis, forensic techniques, and cross-referencing against a stored knowledge base.

---

3. Project Objectives

#	Objective	Status
1	Automatically process identity-document images and PDF files	Planned
2	Identify the type of identity document from its visual layout	Planned
3	Extract structured information fields using OCR	Planned
4	Analyse visual and structural characteristics of documents	Planned

#	Objective	Status
5	Apply image forensic techniques to detect potential digital manipulation	Planned
6	Validate extracted text fields against format rules and cross-document consistency	Planned
7	Detect duplicate documents or document numbers associated with different identities	Planned
8	Generate structured anomaly signals for each detected suspicious indicator	Planned
9	Compute a composite, explainable risk assessment score	Planned
10	Provide supporting evidence for each anomaly	Planned
11	Present results via a web-based interface and REST API	Planned
12	Support export of screening reports for human/official verification	Planned

4. Key Features

All features listed below are **planned**. The repository is currently in the initialization and planning phase.

Feature	Status	Planned Technology	Description
Document Upload (image/PDF)	Planned	FastAPI, Python	Accept JPEG, PNG, PDF uploads via REST API
File Validation	Planned	Python (Pillow, PyMuPDF)	Validate file format, size, and image quality
Image Preprocessing	Planned	OpenCV	Deskewing, denoising, contrast normalization, resizing
Document Type Classification	Planned	CNN / ML classifier	Identify document type
OCR Text Extraction	Planned	Tesseract / PaddleOCR	Extract raw text from document regions
Structured Field Extraction	Planned	Regex, NLP	Parse OCR output into structured fields
Image Forensics	Planned	OpenCV, forensic algorithms	Detect copy-paste, splicing, and re-compression artefacts

Feature	Status	Planned Technology	Description
Tampering Detection	Planned	ML / Deep Learning	Detect evidence of digital editing
Field Validation	Planned	Rule-based engine	Validate field formats and cross-field consistency
Duplicate Detection	Planned	Database lookup, hashing	Identify documents or numbers seen before
Identity Consistency Check	Planned	Database comparison	Flag document numbers linked to different identity fields
Anomaly Detection	Planned	ML / rule-based	Detect and enumerate suspicious signals
Risk Scoring	Planned	Weighted scoring model	Aggregate anomalies into a composite risk level
Explainable AI Report	Planned	Custom reporting engine	Communicate anomalies with supporting evidence
REST API	Planned	FastAPI	Backend API for upload, processing, and result retrieval
Web Frontend	Planned	React / HTML + JS	User-facing interface
Admin Dashboard	Planned	Frontend framework	View processing history and analytics
Docker Containerisation	Planned	Docker, Docker Compose	Container-based deployment

5. System Workflow

The following diagram describes the proposed end-to-end processing workflow:

DIAGRAM (Mermaid) *Render on GitHub for interactive view*

```
flowchart TD
A([Document Upload\nImage or PDF]) --> B[File Validation\nFormat · Size · Quality]
B --> C[Image Preprocessing\nDeskew · Denoise · Normalize]
C --> D[Document Type Classification]
D --> E[OCR Text Extraction\nTesseract or PaddleOCR]
E --> F[Structured Field Extraction\nName · DOB · Document Number]
F --> G[Image Forensics\nCopy-Paste · Splicing · Recompression]
F --> H[Field Consistency Validation\nFormat Rules · Cross-field Logic]
F --> I[Duplicate and Identity Consistency Check\nDatabase Lookup · Hash Matching]
G --> J[Anomaly Aggregation Engine]
H --> J
I --> J
J --> K[Risk Assessment\nWeighted Composite Score]
K --> L[Explainable Screening Report\nAnomaly List · Evidence · Risk Level]
L --> M([Human and Official Verification])
```

Status: This workflow represents the **proposed design**. Implementation has not yet commenced.

6. Important Fraud Scenario

Same Document Number with Different Identity Information

One of the most significant fraud signals this system is designed to detect is the re-use of a genuine document number with different identity fields. The following uses **entirely fictional information**.

Record already in the database:

Field	Value
Document Number	ABC1234567
Name	Rahul Kumar
Date of Birth	12-05-2001

Newly submitted document:

Field	Value
Document Number	ABC1234567
Name	Amit Sharma
Date of Birth	12-05-2001

The document numbers are identical but the names differ — a suspicious identity-consistency signal.

Detection Flow

DIAGRAM (Mermaid) *Render on GitHub for interactive view*

```
flowchart TD
A([New Document Submitted]) --> B[OCR Extraction]
B --> C[Extract Doc No. ABC1234567\nName: Amit Sharma · DOB: 12-05-2001]
C --> D[Database Search by Document Number]
D --> E{Document Number\nFound?}
E -- No --> F[Store as New Record]
E -- Yes --> G[Retrieve Existing Record\nName: Rahul Kumar · DOB: 12-05-2001]
G --> H[Compare Identity Fields]
H --> I{Name Match?}
I -- Yes --> J[No Inconsistency Detected]
I -- No --> K[ANOMALY: Name Mismatch\nABC1234567 linked to two different names]
K --> L[Risk Score Increased]
L --> M([Flag for Human Verification])
```

Important: A name mismatch on a shared document number does **not** automatically prove fraud. It is a suspicious signal requiring additional human and official verification. Possible explanations include data entry errors, OCR misreads, document reissuance, or deliberate misuse.

7. Types of Anomalies

Anomaly	Example	Detection Approach	Planned Response
Text / Field Manipulation	Name changed	Image forensics + OCR comparison	Flag
Date of Birth Modification	DOB changed	OCR + format validation + DB check	Flag
Document Number Modification	Last digits altered	OCR + format rules + DB lookup	Flag
Photograph Replacement	Different person's photo	Computer vision + face analysis	Flag
Image Splicing	Region from another image	Forensic artefact detection (proposed)	Flag
Copy-Paste Manipulation	Text block repositioned	Visual similarity analysis (proposed)	Flag
Font Inconsistency	Different font in edited field	Layout + typography analysis (proposed)	Flag
Layout / Alignment Inconsistency	Text field misaligned	Document structure comparison (proposed)	Flag
Screenshot Substitution	Digital screenshot submitted	Screen artefact detection (proposed)	Flag
Reprint / Recapture	Modified document re-scanned	Image quality analysis (proposed)	Flag
Duplicate Document	Same document for multiple people	Hash matching + embedding similarity (proposed)	Flag
Same ID + Different Name	Doc no. linked to two names	Database comparison	Flag
Same ID + Different DOB	Doc no. linked to two DOBs	Database comparison	Flag
Synthetic / AI-Generated Document	Fully fabricated document	Forensic / ML-based detection (proposed)	Flag
Metadata Inconsistency	Creation date after claimed issue date	File metadata analysis (proposed)	Flag

8. AI / ML Architecture

8.1 Computer Vision (OpenCV)

- **Preprocessing:** Geometric correction, noise reduction, contrast normalisation, adaptive thresholding, edge detection
- **Feature extraction:** Structural analysis of document regions, field boundaries, and security element locations
- **Manipulation analysis:** Statistical noise analysis, block artefact detection, and local variance analysis to identify regions inconsistent with surrounding content

8.2 OCR — Text Extraction

The system proposes **Tesseract OCR** or **PaddleOCR** for text extraction. OCR output is parsed using regular expressions and named entity recognition to populate structured field objects (name, date of birth, document number, etc.).

8.3 Machine Learning

Module	Proposed Approach
Document type classification	CNN-based classifier trained on document layout features
Tampering detection	Binary classifier (authentic vs. manipulated) using forensic image features
Anomaly detection	Isolation Forest or one-class SVM on extracted feature vectors
Risk scoring	Weighted aggregation of anomaly signals with tunable thresholds

8.4 Deep Learning (Proposed)

Depending on dataset availability and project progress, the system may incorporate:

- **Convolutional Neural Networks (CNNs)** for document classification and visual feature extraction
- **Pre-trained Vision Transformers (ViT)** for fine-grained manipulation detection, building on the field-localised approach demonstrated in FLiD (Kumar et al., 2026)
- **MobileNetV3 / EfficientNet** as lightweight backbones for resource-efficient inference

8.5 Explainable AI — Planned

Planned explainability outputs include:

- **Anomaly description:** Plain-language explanation of the detected anomaly
- **Affected field:** Specific document field associated with the anomaly
- **Confidence level:** Model confidence or rule-match strength
- **Supporting evidence:** Highlighted region, field value comparison, or forensic artefact description
- **Risk level:** Low / Medium / High / Critical

> Specific XAI methods (LIME, SHAP, gradient-based saliency maps) will be selected during implementation based on the chosen model architecture.

---

9. Research and Literature Foundation

The following three research papers were independently verified and are directly relevant to the methods and objectives of this project.

Paper 1: MIDV-2020 — A Comprehensive Benchmark Dataset for Identity Document Analysis

Field	Detail
Title	MIDV-2020: A Comprehensive Benchmark Dataset for Identity Document Analysis
Authors	Konstantin Bulatov, Ekaterina Emelianova, Daniil Tropin, Natalya Skoryukina, Yulia Chernyshova, Alexey Sheshkus, Vladimir L. Arlazarov
Year	2022
Journal / Conference	Computer Optics, Vol. 46, No. 2
Publisher	Image Processing Systems Institute (IPSI RAS)
arXiv	arXiv:2107.00396
URL	https://arxiv.org/abs/2107.00396

Problem Addressed:

A key obstacle in identity document analysis research is the scarcity of publicly available, labelled datasets. Existing datasets at the time of publication lacked diversity in document types, capturing conditions, or field-value variability, and were typically designed for only a subset of document analysis tasks.

Dataset and Methodology:

MIDV-2020 consists of **1,000 video clips**, **2,000 scanned images**, and **1,000 photographs** covering 1,000 unique mock identity documents. Each document uses unique synthetic text field values and unique artificially generated face images, with rich annotation covering document localisation, OCR, and face detection. In total, MIDV-2020 contains **72,409 annotated images**.

Key Findings:

MIDV-2020 establishes a large-scale benchmark covering multiple document analysis sub-tasks simultaneously. The use of synthetic, privacy-preserving data allows public release without exposing real personal information.

Limitations:

MIDV-2020 is primarily designed for document recognition tasks rather than specifically for forgery or tampering detection. A derivative dataset, MIDV-DM, has since been introduced for document manipulation detection and localisation, extending MIDV-2020 with 8,000 automatically generated manipulated images covering eight manipulation types.

Relevance to This Project:

MIDV-2020 is a directly applicable resource for training and evaluating document type classification and OCR components. MIDV-DM is particularly relevant for the tampering detection module.

Paper 2: FLiD — Field-Localized Forgery Detection for Digital Identity Documents

Field	Detail
Title	Field-Localized Forgery Detection for Digital Identity Documents

Field	Detail
Authors	Abhishek Kumar, Riya Tapwal, Carsten Maple, Mark Hooper
Year	2026
Journal / Conference	arXiv preprint, cs.CV
arXiv	arXiv:2605.09089
DOI	10.48550/arXiv.2605.09089
URL	https://arxiv.org/abs/2605.09089

Problem Addressed:

Existing image forgery detectors are designed for natural images and do not account for the structured layout of identity documents, where security-relevant information is concentrated in specific semantic fields. Full-document processing is computationally expensive and dilutes the signal because the manipulated area is typically very small (0.3–4% of pixels) relative to the entire document.

Methodology:

FLiD (Field-Localised Forgery Detection) uses a **fine-tuned YOLO11 detector** to localise facial and textual regions, then extracts compact representations using a **frozen MobileNetV3-Small backbone**. A small 191K-parameter classification head makes the final forgery decision. Face and text detectors are trained independently and combined through **score-level fusion** for documents with simultaneous manipulations.

Key Findings:

Under 5-fold cross-validation, FLiD achieves:

- AUC **834** for face region manipulation detection
- AUC **926** for text region manipulation detection
- AUC **837** for combined attack detection

FLiD reduces Equal Error Rate by **28–29 percentage points** compared to a full-document baseline, requires **13x fewer trainable parameters**, and **21x fewer FLOPs** per field.

Limitations:

This is a preprint and has not yet undergone full peer review. Generalisation to highly diverse real-world document collections with varying capture conditions remains to be established.

Relevance to This Project:

FLiD directly validates the proposed capstone approach of combining field localisation, lightweight CNN backbone, and field-level forgery classification. The field-localised paradigm and score-level fusion strategy are applicable design patterns for the tampering detection module.

Paper 3: DocForge-Bench — A Comprehensive Benchmark for Document Forgery Detection and Ana

Field	Detail
Title	DocForge-Bench: A Comprehensive Benchmark for Document Forgery Detection and Analysis

Field	Detail
Year	2026
Journal / Conference	arXiv preprint, cs.CV
arXiv	arXiv:2603.01433
URL	https://arxiv.org/abs/2603.01433

Problem Addressed:

General-purpose image manipulation detection methods assume the tampered region comprises 10–30% of the image area. Real-world identity document forgeries typically affect only **0.3–4% of pixels**, which invalidates standard decision thresholds and evaluation metrics used in natural image manipulation detection.

Methodology:

DocForge-Bench introduces a structured benchmark covering text-region tampering, commercial receipt forgery, and identity document forgery. It systematically evaluates multiple existing detectors (TruFor, CAT-Net, and others) against document-specific datasets including DocTamper, T-SROIE, RealTextManipulation, Tampered-IC13, MixTamper, FantasyID, and FSTS-1.5k, providing a rigorous cross-dataset evaluation framework and taxonomy of document manipulation types.

Key Findings:

Leading general-purpose manipulation detectors fail significantly on identity document forgeries due to extreme class imbalance between authentic and manipulated pixels. Domain-specific models trained with document-appropriate thresholds substantially outperform general detectors on this task.

Limitations:

As a preprint, peer-review conclusions are pending. Dataset licences and accessibility vary across the catalogued datasets.

Relevance to This Project:

DocForge-Bench provides the critical justification for why generic manipulation detectors are insufficient for identity documents, and catalogues the datasets (DocTamper, FantasyID, MIDV-DM) most applicable for training and evaluation in this capstone.

---

10. Research Gap

Gap: Integrated Multi-Modal Identity Screening Pipeline with Explainable Risk Assessment

Existing published systems tend to address individual sub-problems in isolation:

Sub-problem	State of Existing Research
Identity document recognition / OCR	Well-studied (MIDV-2020 and successors)
Full-document forgery classification	Partially addressed (FLiD, DocForge-Bench)
Tampered text-region localisation	Active research area (DocTamper, CAT-Net variants)

Sub-problem	State of Existing Research
Photograph replacement detection	Addressed in face anti-spoofing literature
Identity consistency / duplicate detection	Primarily a systems engineering problem; limited academic ML work
Explainable risk assessment for documents	Very limited published work

The gap this capstone addresses: No publicly documented, integrated, open-source pipeline currently combines all of the following in a single deployable system:

- OCR-based structured field extraction
- Computer vision-based document analysis
- Image forensics for manipulation detection
- Field-level validation and consistency checking
- Cross-document identity consistency verification
- Duplicate detection
- Anomaly aggregation and risk scoring
- Explainable, human-readable screening reports
- A deployable REST API and web interface

This capstone proposes to investigate the feasibility of such an integrated pipeline, evaluate its effectiveness on available public datasets, and document the engineering challenges encountered.

---

11. Dataset Strategy

No dataset has been included in this repository. The project is in the planning phase.

11.1 Publicly Available Datasets Under Consideration

Dataset	Description	Relevance
MIDV-2020	72,409 annotated images of 1,000 unique mock identity documents	Document recognition, OCR baseline
MIDV-DM	MIDV-2020 extended with 8,000 manipulated images (8 manipulation types)	Tampering detection training and evaluation
DocTamper	Document-specific tampering dataset with pixel-level localisation masks	Text-region tampering localisation
FantasyID	Identity document forgery dataset	Identity document forgery classification
FSTS-1.5k	Forged seal/stamp and text dataset	Text manipulation detection
Tampered-IC13	ICDAR 2013 text dataset with synthetic tampering	Text manipulation detection

11.2 Synthetic Data Generation

Given that real identity document images are subject to strict privacy and legal constraints, the project will explore:

- Generating mock identity documents with fictional names, dates, and document numbers
- Applying controlled, programmatic manipulations (text replacement, image splicing, contrast modification) to create labelled training data
- Augmentation techniques to maintain class balance

11.3 Key Considerations

Consideration	Notes
Privacy and Legal	Real identity document images cannot be collected without explicit consent. All datasets will be publicly available research datasets with appropriate licences, or synthetically generated data using fictional information.
Licensing	Dataset licences will be reviewed before use.
Class Imbalance	Addressed through oversampling, class-weighted loss functions, or augmentation.
Data Splits	Standard 70/15/15 train/validation/test split with stratification.
De-identification	Any experimental data derived from real documents must be fully de-identified before use.

12. System Architecture

DIAGRAM (Mermaid) *Render on GitHub for interactive view*

```
graph TB
subgraph FE["Frontend — Planned"]
F1[Document Upload UI]
F2[Result and Report View]
F3[Admin Dashboard]
end

subgraph API["API Layer — Planned FastAPI"]
A1[POST /documents/upload]
A2[GET /documents/id/result]
A3[GET /documents/id/report]
A4[GET /admin/history]
end

subgraph PROC["Document Processing Engine — Planned Python"]
P1[File Validator]
P2[Image Preprocessor OpenCV]
P3[Document Type Classifier CNN]
P4[OCR Engine Tesseract or PaddleOCR]
P5[Field Extractor Regex and NLP]
end

subgraph AI["AI and Forensic Analysis — Planned"]
M1[Image Forensics Module OpenCV]
M2[Tampering Detector CNN and ML]
M3[Duplicate Detector Hashing and Embeddings]
```

```
M4[Identity Consistency Checker Database Comparison]
end

subgraph VAL["Validation and Anomaly Engine — Planned"]
V1[Field Validation Rule-based]
V2[Anomaly Aggregator]
V3[Risk Scoring Engine]
V4[Explainability Module]
end

subgraph DB["Database — Planned PostgreSQL"]
D1[(Documents)]
D2[(ExtractedFields)]
D3[(VerificationResults)]
D4[(Anomalies)]
D5[(ProcessingHistory)]
end

FE --> API
API --> PROC
PROC --> AI
PROC --> VAL
AI --> VAL
VAL --> DB
DB --> API
```

13. Technology Stack

All technologies listed are **planned**. Final choices are subject to change during development.

Category	Technology	Purpose
Language	Python 3.10+	Primary development language
Backend / API	FastAPI	REST API framework
Frontend	React.js / HTML + JavaScript	Web interface
OCR	Tesseract OCR / PaddleOCR	Text extraction from document images
Computer Vision	OpenCV	Image preprocessing, feature extraction, forensic analysis
Machine Learning	scikit-learn	Classical ML models (classification, anomaly detection)
Deep Learning	PyTorch	CNN and transformer-based models
Image Processing	Pillow (PIL)	Image loading and format handling
PDF Processing	PyMuPDF	PDF to image conversion
Database	PostgreSQL	Storage of documents, fields, anomalies, and history
ORM	SQLAlchemy	Database interaction layer

Category	Technology	Purpose
Containerisation	Docker + Docker Compose	Reproducible deployment
Data Processing	NumPy, Pandas	Numerical computation and data manipulation
Testing	pytest	Unit and integration testing
Version Control	Git + GitHub	Source code management

14. Project Structure

The repository is currently empty (initialization phase). The following structure represents the **planned** project layout.

```
AI-Based-Fake-Identity-Document-Screening-System/
|
+-- README.md # Project documentation (this file)
|
+-- backend/ # Planned: Python backend
| +-- main.py # FastAPI application entry point
| +-- requirements.txt # Python dependencies
| +-- .env.example # Environment variable template
| |
| +-- api/ # API route definitions
| | +-- routes/
| | | +-- documents.py # Document upload and status endpoints
| | | +-- admin.py # Admin / history endpoints
| | +-- dependencies.py # FastAPI dependency injection
| |
| +-- core/ # Core configuration and utilities
| | +-- config.py # Application settings
| | +-- database.py # Database connection and session
| | +-- security.py # Authentication utilities
| |
| +-- processing/ # Document processing pipeline
| | +-- validator.py # File validation
| | +-- preprocessor.py # Image preprocessing
| | +-- classifier.py # Document type classification
| | +-- ocr_engine.py # OCR wrapper
| | +-- field_extractor.py # Structured field extraction
| |
| +-- forensics/ # Image forensics and AI analysis
| | +-- image_forensics.py # Copy-paste, splicing detection
| | +-- tampering_detector.py # ML-based tampering detection
| | +-- duplicate_detector.py # Hash and embedding-based duplicate detection
| |
| +-- validation/ # Field validation and anomaly detection
| | +-- field_validator.py # Rule-based field validation
| | +-- identity_consistency.py # Identity consistency check
| | +-- anomaly_aggregator.py # Anomaly aggregation
| | +-- risk_engine.py # Risk score computation
| | +-- explainability.py # Explainable AI report generation
| |
| +-- models/ # Database ORM models
| | +-- document.py
```

```
| | +-- extracted_field.py
| | +-- verification_result.py
| | +-- anomaly.py
| | +-- processing_history.py
| |
| +-- tests/ # Backend tests
| +-- test_api.py
| +-- test_ocr.py
| +-- test_forensics.py
| +-- test_validation.py
|
+-- frontend/ # Planned: Web frontend
| +-- index.html
| +-- src/
| | +-- components/
| | +-- pages/
| +-- package.json
|
+-- ml/ # Planned: ML model training
| +-- notebooks/ # Research and exploration notebooks
| +-- training/
| | +-- train_classifier.py
| | +-- train_tampering_detector.py
| +-- models/
| +-- evaluation/
| +-- evaluate.py
|
+-- data/ # Planned: Dataset management (gitignored)
| +-- raw/
| +-- processed/
| +-- synthetic/
|
+-- docker/ # Planned: Docker configuration
| +-- Dockerfile.backend
| +-- Dockerfile.frontend
| +-- docker-compose.yml
|
+-- docs/ # Planned: Extended documentation
| +-- architecture.md
| +-- api_reference.md
| +-- screenshots/
|
+-- .gitignore
```

15. Installation

> **Note:** The repository is currently empty. Installation instructions describe the **planned** setup procedure.

Prerequisites

- Python10 or higher
- Node.js 18+ (for frontend)
- PostgreSQL 14+
- Docker and Docker Compose (optional)

- Tesseract OCR (system dependency)
- Git

Step 1: Clone the Repository

BASH

```
git clone https://github.com/sumitkumarbarnwal/AI-Based-Fake-Identity-Document-Screening-System.git
cd AI-Based-Fake-Identity-Document-Screening-System
```

Step 2: Backend Setup (Planned)

BASH

```
cd backend

# Create and activate a virtual environment
python -m venv venv
source venv/bin/activate # Linux / macOS
venv\Scripts\activate # Windows

# Install Python dependencies
pip install -r requirements.txt
```

Step 3: Install Tesseract OCR (Planned)

BASH

```
# Ubuntu / Debian
sudo apt-get install tesseract-ocr

# macOS (Homebrew)
brew install tesseract

# Windows: Download installer from https://github.com/UB-Mannheim/tesseract/wiki
```

Step 4: Configure Environment Variables (Planned)

BASH

```
cp .env.example .env
# Edit .env with your configuration — see Section 16
```

Step 5: Database Setup (Planned)

BASH

```
createdb ai_document_screening
python -m alembic upgrade head
```

Step 6: Frontend Setup (Planned)

BASH

```
cd ../frontend
npm install
```

Step 7: Run the Application (Planned)

BASH

```
# Backend (from backend/)
uvicorn main:app --reload --host 0.0.0.0 --port 8000

# Frontend (from frontend/, separate terminal)
npm run dev
```

Alternative: Docker (Planned)

BASH

```
docker-compose up --build
```

---

16. Environment Variables

`.env.example` will be provided when the backend is implemented. Never commit actual secrets.

ENV

```
# Application
APP_ENV=development
APP_HOST=0.0.0.0
APP_PORT=8000
SECRET_KEY=your_secret_key_here

# Database
DATABASE_URL=postgresql://user:password@localhost:5432/ai_document_screening

# OCR
TESSERACT_CMD=/usr/bin/tesseract

# File Upload
MAX_UPLOAD_SIZE_MB=10
ALLOWED_EXTENSIONS=jpg,jpeg,png,pdf

# ML Models
MODEL_DIR=./ml/models

# Storage
UPLOAD_DIR=./uploads

# Logging
LOG_LEVEL=INFO
```

---

17. API Documentation

Note: The backend API has not yet been implemented. The following describes the **planned** REST API.

Base URL: `http://localhost:8000/api/v1`

Planned Endpoints

Endpoint	Method	Purpose
`/documents/upload`	POST	Upload a document image or PDF for screening
`/documents/{id}/status`	GET	Check processing status of a submitted document
`/documents/{id}/result`	GET	Retrieve the screening result and anomaly list
`/documents/{id}/report`	GET	Download the full explainable screening report
`/admin/documents`	GET	List all processed documents (admin only)
`/admin/documents/{id}`	DELETE	Remove a document record (admin only)
`/health`	GET	API health check

Example Request: Document Upload (Planned)

HTTP

```
POST /api/v1/documents/upload
Content-Type: multipart/form-data

file: <document image or PDF>
document_type_hint: "aadhaar"
```

Example Response: Screening Result (Planned)

JSON

```
{
"document_id": "d_abc123",
"status": "completed",
"risk_level": "HIGH",
"risk_score": 78,
"extracted_fields": {
"name": "Amit Sharma",
"date_of_birth": "12-05-2001",
"document_number": "ABC1234567"
},
"anomalies": [
{
"anomaly_id": "a_001",
"type": "IDENTITY_CONSISTENCY",
"description": "Document number ABC1234567 is already associated with a different name in the
database.",
"severity": "HIGH",
"evidence": {
"existing_name": "Rahul Kumar",
```

```
"submitted_name": "Amit Sharma",
"field": "name"
}
}
],
"recommendation": "Flag for human verification",
"processing_time_ms": 1842
}
```

18. Database Design

Status: Proposed. No database has been implemented.

DIAGRAM (Mermaid) *Render on GitHub for interactive view*

```
erDiagram
DOCUMENTS {
uuid id PK
string filename
string file_type
string storage_path
string document_type
timestamp uploaded_at
string status
}

EXTRACTED_FIELDS {
uuid id PK
uuid document_id FK
string field_name
string field_value
float ocr_confidence
timestamp extracted_at
}

VERIFICATION_RESULTS {
uuid id PK
uuid document_id FK
string risk_level
int risk_score
string recommendation
timestamp created_at
}

ANOMALIES {
uuid id PK
uuid document_id FK
string anomaly_type
string description
string severity
json evidence
timestamp detected_at
}

PROCESSING_HISTORY {
uuid id PK
```

```
uuid document_id FK
string stage
string status
string message
timestamp recorded_at
}

DOCUMENTS ||--o{ EXTRACTED_FIELDS : "has"
DOCUMENTS ||--o| VERIFICATION_RESULTS : "produces"
DOCUMENTS ||--o{ ANOMALIES : "generates"
DOCUMENTS ||--o{ PROCESSING_HISTORY : "tracked by"
```

19. Screenshots

No screenshots are currently available. The application interface has not yet been implemented.

Screenshots will be added as the application reaches the prototype stage. Planned screenshots include:

- Document upload interface
- OCR extraction result view
- Anomaly detection report
- Risk assessment dashboard
- Explainable screening report

20. Testing Strategy

Status: Testing has not yet been implemented.

Test Type	Tool	Description
Unit Tests	pytest	Test individual functions in preprocessing, OCR, field extraction, validation, and risk scoring modules
API Tests	pytest + httpx	Test each REST endpoint for correct request handling, status codes, and response schemas
OCR Tests	pytest	Evaluate OCR accuracy on a set of labelled test document images
Model Tests	pytest	Validate ML model loading, inference, and output format
Integration Tests	pytest	Test the full processing pipeline end-to-end with sample documents
Security Tests	Manual + tools	Test for file upload vulnerabilities, path traversal, and API authentication
Performance Tests	locust / pytest-benchmark	Measure processing time and throughput under concurrent load

Test Type	Tool	Description
Adversarial Tests	Manual	Test the system against deliberately manipulated documents to assess detection robustness

21. Model Evaluation

Status: No ML models have been implemented or trained. No numerical results are reported.

Planned Evaluation Metrics

Metric	Description
Accuracy	Overall classification correctness (authentic vs. manipulated)
Precision	Proportion of flagged documents that are genuinely suspicious
Recall	Proportion of genuinely suspicious documents correctly flagged
F1-Score	Harmonic mean of precision and recall
ROC-AUC	Area under the receiver operating characteristic curve
False Positive Rate	Rate of authentic documents incorrectly flagged
False Negative Rate	Rate of manipulated documents incorrectly cleared
OCR Field Accuracy	Percentage of text fields correctly extracted and parsed
Processing Time	Average end-to-end processing time per document

False negatives (failing to detect manipulation) and false positives (incorrectly flagging authentic documents) have different real-world consequences. Evaluation will consider this tradeoff explicitly in the context of a human-in-the-loop screening workflow.

22. Security and Privacy

Because this project processes identity documents — some of the most sensitive personal data — security and privacy are design-level requirements.

Control	Status	Notes
No hard-coded secrets	Design requirement	All credentials managed via environment variables
Secure environment variables	Planned	`.env` files excluded from version control via `.gitignore`

Control	Status	Notes
Input validation	Planned	File type, size, and format validation before processing
Upload size limits	Planned	Configurable maximum upload size
Authentication	Planned	Token-based API authentication for all non-public endpoints
Access control	Planned	Admin-only endpoints for history and document management
Data minimisation	Design requirement	Only the minimum data necessary for screening will be stored
Encrypted storage	Planned	Document images stored with encryption at rest
Secure logging	Planned	Logs must not contain raw extracted personal data
Data retention	Planned	Documents and fields deleted after a defined retention period
Synthetic/de-identified data	Design requirement	Only publicly licensed or synthetically generated data used in development
HTTPS	Planned	TLS encryption for all API communication in production

Important Privacy Note

In a production context, applicable data protection regulations (such as the DPDP Act 2023 in India or GDPR in the EU) would govern data collection, storage, retention, and subject rights. During development and research, only publicly available research datasets with appropriate licences, or synthetically generated data using fictional information, will be used.

23. Limitations

Limitation	Description
OCR errors	OCR accuracy degrades with poor image quality, unusual fonts, or low-contrast documents
Image quality dependency	The system's effectiveness depends heavily on the quality of the submitted image
Unseen manipulation techniques	A model trained on known techniques may fail to detect novel forgery methods
Dataset limitations	Publicly available identity document datasets are limited in diversity
False positives	The system may flag authentic documents as suspicious

Limitation	Description
False negatives	The system may fail to detect sophisticated or novel manipulations
No legal authority	Automated screening does not constitute legal verification
Generalisation	Performance may vary significantly across document types, issuing authorities, or countries
Privacy constraints	Use of real identity documents for training is subject to strict legal and ethical constraints
Decision-support only	This system is designed to support — not replace — human review and official verification

24. Roadmap

Phase 1: Research and Planning (Months 1–2)

- ■ Repository initialized
- ■ Literature review (papers identified — see Section 9)
- ■ Dataset research and licensing review
- ■ Requirements specification
- ■ System architecture design
- ■ Technology stack finalisation
- ■ Project structure setup

Phase 2: Document Processing Foundation (Months 2–3)

- ■ Backend project scaffolding (FastAPI)
- ■ Document upload API endpoint
- ■ File validation module
- ■ Image preprocessing pipeline (OpenCV)
- ■ OCR integration (Tesseract / PaddleOCR)
- ■ Structured field extraction
- ■ Database schema implementation
- ■ Unit tests for processing modules

Phase 3: AI and ML Components (Months 3–4)

- ■ Document type classification model
- ■ Image forensics module
- ■ Tampering / manipulation detection model

- Model training on available datasets
- Model evaluation and benchmarking

Phase 4: Identity Consistency and Validation (Month 4)

- Field validation rule engine
- Database-based identity consistency checking
- Duplicate document detection
- Anomaly aggregation engine
- Risk scoring model

Phase 5: Application Layer (Months 4–5)

- Complete REST API implementation
- Web frontend: document upload interface
- Web frontend: result and anomaly display
- Explainable AI reporting module
- Admin dashboard
- API authentication

Phase 6: Deployment, Evaluation, and Documentation (Months 5–6)

- Docker containerisation
- Full integration testing
- Performance evaluation and optimisation
- Security audit
- Adversarial testing
- Deployment (staging environment)
- Final project documentation and report
- Capstone presentation materials

---

25. Responsible AI Disclaimer

This system is designed for preliminary document screening and academic research purposes.

An automated risk signal generated by this system **does not** establish that a document is forged, that an identity is fraudulent, or that any individual has engaged in illegal activity. Automated systems can produce false positives and false negatives. No automated screening system can authenticate a document in a legal sense.

All high-risk signals must be reviewed by appropriately qualified human reviewers and, where applicable, by official verification authorities before any consequential decision is made.

This system **must not** be used to:

- Make irreversible decisions about individuals solely on the basis of an automated prediction
- Deny services, access, or rights to individuals without human review
- Replace official document verification processes required by law

The developers of this system take no responsibility for decisions made on the basis of automated outputs without appropriate human oversight.

---

26. Contributing

Contributions are welcome as the project develops. To contribute:

- **Fork** the repository on GitHub
- **Create a branch** for your feature or fix:

  BASH

  ```
  git checkout -b feature/your-feature-name
  ```

- **Make your changes** following the project's code style and documentation conventions
- **Write or update tests** for your changes
- **Commit** with a clear message:

  BASH

  ```
  git commit -m "feat: add image preprocessing pipeline"
  ```

- **Push** to your fork:

  BASH

  ```
  git push origin feature/your-feature-name
  ```

- **Open a Pull Request** with a clear description of your changes

Please ensure all tests pass before submitting a pull request. For significant architectural changes, open an issue for discussion before starting development.

---

27. License

License: Not yet specified.

No `LICENSE` file is currently present in this repository. The license will be added during the project development phase. Until specified, all rights are reserved by the repository owner.

---

28. Author

Sumit Kumar Barnwal

- **GitHub:** github.com/sumitkumarbarnwal
- **Repository:** AI-Based-Fake-Identity-Document-Screening-System

---

This README was prepared for academic and portfolio purposes as part of a 6-month capstone project. All identity information used in examples is entirely fictional. This system is designed as a research and decision-support tool, not as an authoritative fraud detection system.

This PDF was generated from the project README.md.
Visit the GitHub repository for the latest version:
github.com/sumitkumarbarnwal/AI-Based-Fake-Identity-Document-Screening-System